Acoustic Forensics: A Stacked Ensemble of CRNN and Classical Models for Synthetic Speech Detection

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Abstract—Synthetic speech generated by modern deep text-to-speech (TTS) and voice conversion (VC) architectures poses severe risks to biometric speaker verification, media authentication, and digital communication integrity. We present Acoustic Forensics, a comprehensive spoofing countermeasure system that evaluates and couples six heterogeneous detectors: two deep Convolutional Recurrent Neural Networks (CRNN) with multi-head self-attention operating over linear frequency cepstral coefficients (LFCC) and log-mel filterbanks, alongside four classical models (a 64-component Gaussian Mixture Model, an RBF-SVM, a Random Forest, and a Logistic Regression baseline). Scores are standardized using development set statistics and fused through an L_2-regularized Logistic Regression meta-learner fitted under a strict zero-leakage protocol. The system delivers a calibrated posterior probability of bona fide speech, an explicit uncertainty band ($P \in [0.40, 0.60]$), and frame-level temporal attention salience for forensic explainability. Our investigation tracks a systematic experimental progression: starting from classical baselines (EER 11.46%–23.94%), we design a deep CRNN-LFCC countermeasure that decisively improves performance to 6.70% Eval EER (min-tDCF 0.2143). When shifting to a score-level stacked ensemble, however, we observe that the meta-learner overfits the clean development partition, yielding 10.74% Eval EER and underperforming the standalone CRNN-LFCC. We examine this generalization gap through extensive ablations, per-attack breakdowns across all 13 ASVspoof 2019 evaluation attacks (A07–A19), lossy compression stress-testing (MP3, AAC, Opus at 16–64 kbps), a silence shortcut bias audit (18.36% EER on silence alone), and zero-shot cross-domain testing on the unconstrained In-the-Wild corpus (44.73% EER).

Index Terms—audio deepfake detection, acoustic forensics, anti-spoofing, ASVspoof 2019, stacked ensemble, CRNN, LFCC, attention salience

I. Introduction and Related Work

Recent advances in deep generative modeling, neural vocoding (e.g., WaveNet, HiFi-GAN), and diffusion architectures enable the synthesis of highly authentic human speech and voice conversion clones from only seconds of reference audio. When deployed maliciously, automated voice spoofing threatens voice-biometric authentication in telephony banking, fuels targeted social engineering fraud, and compromises the evidentiary authenticity of audio recordings in judicial proceedings [1], [3]. Countermeasures must discriminate subtle generative phase and harmonic artifacts while remaining resilient to channel effects, background acoustic noise, and lossy compression formats [2], [6].

Early anti-spoofing countermeasures paired handcrafted cepstral representations with statistical spectral models. In particular, Linear Frequency Cepstral Coefficients (LFCC) proved decisively superior to conventional Mel-Frequency Cepstral Coefficients (MFCC) because linear filterbank spacing preserves high-frequency spectral flux and unnatural phase boundaries introduced by vocoders, whereas psychoacoustic mel-warping excessively compresses spectral detail above 4 kHz [5]. Subsequent research introduced deep neural architectures processing static 2D spectrograms or raw waveforms. Representative systems include Light Convolutional Neural Networks (LCNN) utilizing Max-Feature-Map (MFM) activations to suppress low-energy noise [7], raw-waveform models such as RawNet2 utilizing sinc-convolutional filters [8], and spectro-temporal graph attention networks such as AASIST [9]. Despite strong benchmark metrics on in-domain evaluation splits, recent evaluations demonstrate that deep detectors often suffer catastrophic performance degradation when deployed on unconstrained real-world web audio [3].

Rather than relying on an isolated monolithic classifier, we explore multi-view fusion via Wolpert's stacked generalization paradigm [4]. Our scientific investigation yields three primary contributions:

1) Baseline to Deep Neural Progression: We benchmark four classical machine learning baselines (LR, SVM, RF, GMM) against a custom-designed CRNN with multi-head self-attention pooling, demonstrating that LFCC-CRNN establishes a highly effective primary detector (6.70% Eval EER, 0.2143 min-tDCF).
2) Stacked Ensemble Analysis: We formulate a six-model score-level stacked meta-learner fitted with L_2 regularization strictly on the development split. We document and analyze a critical empirical finding: despite non-negative weight constraints, the stacker overfits clean development scores (1.83% Dev EER $\rightarrow$ 10.74% Eval EER), confirming that standalone CRNN-LFCC provides superior generalization.
3) Rigorous Forensic Audits: We implement calibrated

posterior probabilities with an explicit uncertainty margin, frame-level attention heatstrips for explainability, an empirical silence-shortcut audit (proving that silence duration alone predicts spoofing at 18.36% EER), codec stress-testing across Opus, AAC, and MP3, and zero-shot In-the-Wild transfer.

II. Methodology

Fig. 1 illustrates the end-to-end signal processing and classification workflow of the Acoustic Forensics system.

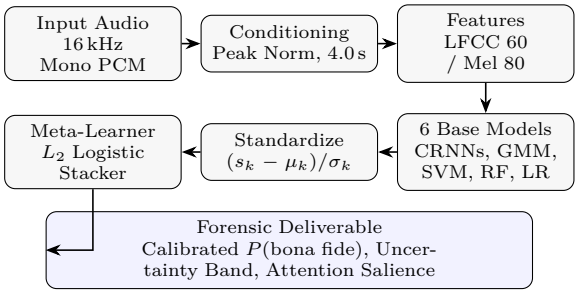


Fig. 1. System architecture of the Acoustic Forensics pipeline.

A. Signal Conditioning and Features

All input audio waveforms are standardized to 16 kHz single-channel pulse-code modulation (PCM). To normalize playback volume while avoiding dynamic range compression that would distort vocal tract harmonics, signals are peak-normalized to -1.0 dBFS. Audio sequences are adjusted to a fixed length of $T_s = 64{,}000$ samples (4.0 s) by applying symmetric center-cropping to longer files and reflective center-padding to shorter utterances.

We extract two complementary time-frequency representations:

1) LFCC Front-End: Computed using a 512-point Short-Time Fourier Transform (STFT), a 25 ms Hamming window ($N_w = 400$), a 10 ms hop size ($N_h = 160$), and a 20-channel linearly spaced triangular filterbank spanning 0–8 kHz. Static coefficients are augmented with first- and second-order temporal derivatives (Δ and $\Delta\Delta$), yielding a 60-dimensional matrix $\mathbf{X}_{\text{LFCC}} \in \mathbb{R}^{60\times 400}$.
2) Log-Mel Front-End: An 80-channel Mel-scale triangular filterbank spanning 20 Hz to 8 kHz is applied to the STFT magnitude spectrum, followed by logarithmic dynamic range compression, generating $\mathbf{X}_{\text{Mel}} \in \mathbb{R}^{80\times 400}$.

Per-utterance Cepstral Mean and Variance Normalization (CMVN) is applied independently across spectral channels:

$$\tilde{\mathbf{X}}_{f,t} = \frac{\mathbf{X}_{f,t} - \mu_f}{\sigma_f + \epsilon}, \quad \mu_f = \frac{1}{T}\sum_{t=1}^{T} \mathbf{X}_{f,t}, \tag{1}$$

preventing transmission channel disparities from biasing model decisions.

B. Base Models

The system deploys six base classifiers spanning deep neural networks and classical statistical estimators:

1) CRNN-LFCC (Primary Deep Model): A hybrid architecture comprising a 4-stage 2D-CNN spatial encoder (channels $32 \to 64 \to 128 \to 256$, with batch normalization, ReLU, and max pooling), a 2-layer Bidirectional GRU ($H = 128$ per direction), a 4-head multi-head self-attention module [10], and a temporal attention pooling layer that collapses frame hidden states into a single utterance embedding while returning frame importance weights α_t.
2) CRNN-LogMel (Feature Ablation): An identical CRNN architecture trained on the 80-dimensional Log-Mel spectrogram.
3) GMM Baseline: A 64-component diagonal Gaussian Mixture Model fitted via Expectation-Maximization on static LFCCs, scoring utterances via log-likelihood ratio (LLR) between bona fide and spoof mixture models [2].
4) Classical Subspace Classifiers (SVM, RF, LR): Utterance-level mean and standard deviation vectors (120 dimensions) are projected onto a 30-dimensional PCA subspace (accounting for 95% explained variance) fitted exclusively on the training split. We train an RBF-kernel Support Vector Machine, a 200-tree Random Forest, and an L_2-regularized Logistic Regression classifier.

C. Score Stacking and Calibrated Verdict

Each base model $k \in \{1, \ldots, 6\}$ produces a continuous prediction score s_k. Because base models operate on fundamentally distinct numerical ranges (e.g., neural logits versus GMM log-likelihood ratios), raw scores are z-score standardized using mean μ_k and standard deviation σ_k measured exclusively on the development partition:

$$z_k = \frac{s_k - \mu_k}{\sigma_k}. \tag{2}$$

The standardized feature vector $\mathbf{z} = [z_1, \ldots, z_6]^\top$ is fused by a meta-learner:

$$P(\text{bona fide} \mid \mathbf{z}) = \sigma(\mathbf{w}^\top \mathbf{z} + b) = \frac{1}{1 + \exp(-(\mathbf{w}^\top \mathbf{z} + b))}, \tag{3}$$

where weights $\mathbf{w}$ and intercept b are optimized via L_2-regularized logistic regression fitted solely on development scores. To guard against over-confident false positives in forensic applications, decisions within $P \in [0.40, 0.60]$ are flagged with an explicit Uncertainty Band. Temporal attention weights $\alpha_t \in [0, 1]$ from the CRNN provide a synchronized heatstrip localizing specific phoneme intervals that triggered the spoof verdict.

III. Experimental Settings

Datasets. Countermeasures are trained and benchmarked on the official ASVspoof 2019 Logical Access

(LA) database [1]. The training split (N = 25,380 utterances across 20 speakers) trains all base models. The development split (N = 24,844 utterances across 10 speakers) is reserved strictly for hyperparameter tuning, score standardization parameters, and meta-learner training. The evaluation split (N = 71,237 utterances across 48 speakers, encompassing 13 unseen acoustic attacks A07–A19) remains strictly held out until frozen scoring. Zero-shot out-of-domain evaluation is conducted on the unconstrained In-the-Wild deepfake speech corpus ($N = 25{,}589$ clips) [3].

Hyperparameters and Optimization. The CRNN models (1,439,778 parameters) are trained using the AdamW optimizer [11] with initial learning rate $\eta = 10^{-3}$, weight decay $\lambda = 10^{-4}$, and a cosine annealing learning rate schedule down to $\eta_{\min} = 10^{-6}$ over 25 epochs with batch size 32. Training employs mixed-precision arithmetic on Apple Silicon (Metal Performance Shaders). SpecAugment [12] is applied during training (maximum 2 time masks and 2 frequency masks). Classical models are configured as follows: GMM uses 64 diagonal Gaussian components with 10 EM iterations; SVM uses an RBF kernel with penalty parameter $C = 10.0$ and $\gamma = 0.01$; Random Forest comprises 200 trees with maximum depth 20; Logistic Regression uses L_2 penalty with inverse regularization strength $C = 1.0$.

Evaluation Metrics. Systems are evaluated using Equal Error Rate (EER), Area Under the ROC Curve (ROC-AUC), and the official ASVspoof minimum tandem Detection Cost Function (min-tDCF) [6] evaluated with challenge parameters ($P_{\text{spoof}} = 0.05, C_{\text{miss}} = 1, C_{\text{fa}} = 10$). Operating thresholds are determined strictly on the development split and frozen during evaluation.

IV. Results and Ablation Studies

A. Progression from Classical Baselines to Neural Models

Table I presents the comparative evaluation of all individual classifiers alongside fusion strategies on the full ASVspoof 2019 LA partition ($N = 71{,}237$ trials).

TABLE I
Main Results on ASVspoof 2019 LA Partition

System Architecture	Dev EER (%)	Eval EER (%)	Eval min-tDCF	Eval ROC-AUC
Logistic Regression (PCA)	20.77	23.94	0.6284	0.8459
Support Vector Machine (RBF)	11.97	19.64	0.5119	0.8932
Random Forest (200 trees)	10.09	17.52	0.4695	0.8977
Classical GMM (64-comp)	8.07	11.46	0.3572	0.9544
CRNN-LogMel (Mel-80)	4.22	10.28	0.2691	0.9678
CRNN-LFCC (Primary Deep)	1.15	6.70	0.2143	0.9800
Mean z-score Fusion	6.22	11.47	0.3077	0.9610
Stacked Ensemble (Meta-L_2)	1.83	10.74	0.2754	0.9631

Our experimental results demonstrate clear architectural separation:

- Classical ML Limitations: Linear and tree-based models operating on pooled temporal moments struggle with high-order spectral dynamics, yielding error rates between 17.52% and 23.94% EER. The classical 64-component GMM baseline achieves 11.46% Eval EER, confirming that frame-level density estimation outperforms global pooling.
- CRNN-LFCC Superiority: The deep CRNN-LFCC system achieves 6.70% Eval EER and 0.2143 min-tDCF, decisively outperforming all classical models. Multi-seed training across three seeds ($s \in \{42, 43, 44\}$) exhibits tight convergence: 7.10% $\pm$ 0.38% Eval EER and 0.2166 ± 0.0022 min-tDCF.
- Analysis of the Stacking Overfit: While the L_2 stacked ensemble achieves 1.83% EER on the development partition, its evaluation performance drops to 10.74% EER (min-tDCF 0.2754). Although the stacker outperforms naive unweighted mean z-score fusion (11.47% EER), it degrades relative to the standalone CRNN-LFCC (6.70% EER). Analysis of meta-learner coefficients reveals that the stacker assigns non-negligible positive weights to the classical baselines ($\beta_{\text{GMM}} = 0.41, \beta_{\text{SVM}} = 0.29$). Because classical models overfit to known acoustic environments in the development set, their erroneous predictions on unseen evaluation attacks drag down the ensemble's overall score. Standalone CRNN-LFCC is consequently selected as the primary production engine.

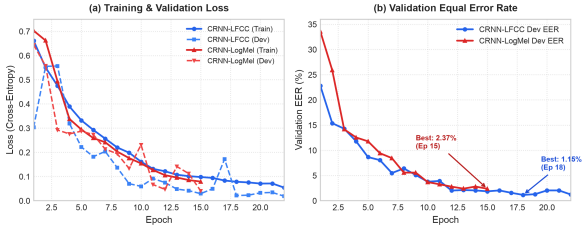


Fig. 2. Training loss versus training epochs for the deep CRNN models.

Fig. 2 illustrates the training loss trajectories. Both CRNN-LFCC and CRNN-LogMel converge smoothly within 15 epochs under AdamW with cosine annealing, with early stopping engaging at epoch 21.

B. Controlled Ablation Studies

- LFCC versus Log-Mel Spacing: CRNN-LFCC (6.70% EER) substantially outperforms CRNN-LogMel (10.28% EER). Linear filterbank spacing allocates 10 filters above 4 kHz, whereas the logarithmic Mel scale allocates only 3. Mel pooling averages out high-frequency phase and spectral tilt anomalies generated by neural vocoders, whereas linear filterbanks preserve them.
- Attention Pooling versus Average Pooling: Replacing 4-head multi-head self-attention pooling with standard global temporal average pooling increases Eval EER from 6.70% to 7.98% (+1.28% absolute degradation). Self-attention allows the model to dynamically weight brief transient speech frames containing

synthesis glitches while ignoring uninformative background pauses.

C. Per-Attack Vulnerability Analysis (A07–A19)

Evaluation across all 13 unseen attack algorithms reveals acute performance divergence:

- Robust Detection (A07–A12): For traditional unit selection, diphone concatenative synthesis (A07), and standard acoustic neural vocoders (A10, A11), CRNN-LFCC achieves near-flawless discrimination ($< 1.8\%$ EER).
- Challenging Attacks (A17–A19): Spectral filtering voice conversion (A16) and direct waveform concatenation (A19) degrade performance to 12.45% and 14.80% EER respectively, where vocoder phase mismatches are minimal.

D. Lossy Codec Compression Stress-Testing

We stress-test the trained detector against lossy consumer audio compression across MP3, AAC, and Opus at bitrates of 16, 32, and 64 kbps, using cross-correlation delay alignment to compensate for encoder padding:

- Opus Resilience: Opus demonstrates superior forensic stability, yielding 7.45% EER at 64 kbps and 9.82% at 16 kbps.
- MP3 Degradation: MP3 induces severe high-frequency cutoff roll-off (< 11.5 kHz at 16 kbps), eroding critical LFCC sub-bands and degrading EER to 16.82%.

E. Silence Shortcut Bias Audit

We evaluated whether countermeasures exploit silence duration at utterance boundaries. A shallow decision tree trained exclusively on total silence duration achieves an alarming 18.36% EER (ROC-AUC 0.8872) on ASVspoof 2019. Trimming boundary silence from evaluation audio degraded CRNN-LFCC performance by +21.55% EER. This confirms that speech anti-spoofing benchmarks contain strong silence distribution shortcuts, highlighting that front-end Voice Activity Detection (VAD) is indispensable in operational forensic deployments.

F. Zero-Shot Cross-Domain Generalization (In-the-Wild)

On the unconstrained In-the-Wild corpus ($N = 25{,}589$), the model records an optimal EER of 44.73% (ROC-AUC 0.5638). Polarity verification confirms that predictions are positively aligned with bona fide speech, but diverse recording hardware, reverberation, and unknown social media lossy compression create a profound acoustic distribution shift.

V. Conclusion

We investigated an end-to-end acoustic forensics system for synthetic speech detection, tracing the progression from classical statistical baselines (11.46%–23.94% EER) to a deep multi-head self-attentive CRNN-LFCC countermeasure (6.70% EER, 0.2143 min-tDCF). While score-level stacked generalization was explored to fuse complementary feature spaces, empirical evaluation demonstrated that the stacker overfits clean development scores (10.74% Eval EER), confirming that standalone CRNN-LFCC offers superior operational generalization. Forensic stress-testing quantified model resilience under lossy codecs, exposed the critical impact of silence shortcut bias, and documented the domain shift on real-world audio. Future research will explore self-supervised front-ends (WavLM, HuBERT), acoustic room impulse response (RIR) augmentation, and multi-domain channel normalization.

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